A Multi-Domain Physics-Informed Neural Network Framework for Sparse-Data Scientific Inference Across Five PDE Systems

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Abstract

Physics-informed neural networks (PINNs) embed governing differential equations directly into the loss function of a neural network, enabling physically consistent predictions even when observational data are sparse. Most published PINN implementations address a single equation in isolation. This paper presents a modular, domain-agnostic framework in which one shared training engine supports five physically distinct PDE systems: 1D Saint-Venant open-channel flow, heat diffusion, wave propagation, two-body Newtonian orbital mechanics, and Euler-Bernoulli beam bending. Each domain is verified independently against either closed-form analytical solutions or conservation-law checks not used in training. A dual-head network architecture is introduced for the coupled Saint-Venant system, together with a gradient-norm adaptive loss-balancing scheme (Wang et al., 2021) that eliminates the need for manual physics-weight tuning. Epistemic uncertainty is quantified via Monte Carlo Dropout (Gal and Ghahramani, 2016) across all domains. A sparse-data benchmark compares PINN against an identical-capacity vanilla MLP at data fractions from 5% to 100%. For the orbital mechanics domain, the PINN reconstructs a complete elliptical orbit from only 18% of the trajectory, achieving $R^2 = 0.87$ on the x-coordinate via Newton's gravitational constraint. For beam elasticity, the framework recovers full deflection curves from 15% of sensor readings at $R^2 = 0.9998$. Persistent limitations in the stiff Saint-Venant momentum equation are reported honestly and traced to gradient-conflict dynamics rather than data insufficiency. An interactive Dash application and a standalone Three.js orbital visualisation accompany the codebase.

1. Introduction

A fundamental tension exists in scientific computing: the equations governing physical systems are often well-known, yet measurements of those systems are sparse, noisy, and expensive. Traditional numerical solvers — finite-difference, finite-element, spectral methods — require dense spatial discretisation regardless of observation density. Data-driven machine learning can approximate solutions from observations alone but typically ignores the underlying physics, leading to predictions that violate conservation laws and fail to extrapolate beyond the training regime.

Physics-informed neural networks, introduced by Raissi, Perdikaris, and Karniadakis (2019), resolve this tension by incorporating the governing partial differential equation directly into the training objective. A PDE residual, computed via automatic differentiation through the network, is added to the data-fit loss.

The network is therefore penalised for violating physics at collocation points scattered across the domain — points that require no measurements at all. The result is a model that learns from sparse data while remaining physically consistent everywhere.

The existing PINN literature is dominated by single-domain implementations. A researcher studying heat conduction rewrites the training loop from scratch when shifting to fluid mechanics; one studying elasticity builds an entirely separate codebase for orbital dynamics. This fragmentation obscures a deeper architectural truth: the network, the optimizer, and the training procedure are identical across all these problems. Only the residual function changes.

This work makes three contributions. First, it introduces a **domain-agnostic PDEModule abstraction** in which any governing equation can be registered as a self-contained class and immediately used with a single shared training engine. Five physically distinct PDE systems are implemented and independently verified. Second, it presents a **dual-head architecture with gradient-norm adaptive loss balancing** for the Saint-Venant shallow-water equations — a stiff, coupled system where naive joint training causes gradient conflict between the two output variables. Third, it provides a **systematic evaluation of epistemic uncertainty** across all five domains via Monte Carlo Dropout, producing per-prediction confidence intervals that are directly actionable for sensor placement and risk assessment.

The work is motivated by a concrete applied problem: hydrological modelling in data-sparse regions. The Volta River basin in Ghana, for example, is monitored by a small number of gauge stations relative to its spatial extent. A physics-informed model that can extrapolate physically from sparse gauge measurements offers real value for flood risk mapping and water resource management in exactly the regime where purely data-driven methods are least reliable.

2. Related Work

The foundational PINN formulation of Raissi et al. (2019) demonstrated solutions to forward and inverse PDE problems including Schrodinger, Burgers, and Navier-Stokes equations. Cai et al. (2021) reviewed PINN applications to fluid mechanics specifically, cataloguing success on laminar pipe flow, arterial blood flow, and lid-driven cavity problems while noting persistent difficulties with turbulence and stiff source terms.

Gradient pathologies in PINN training were characterised systematically by Wang, Teng, and Perdikaris (2021), who showed that the physics loss gradient can dominate the data loss gradient by orders of magnitude early in training, causing the network to prioritise physical consistency over observation fit. They proposed a gradient-norm adaptive weighting scheme in which the physics weight is updated at each step to maintain balance between the two loss components. This is the scheme adopted in the present work.

Uncertainty quantification in PINNs has received comparatively less attention. Gal and Ghahramani (2016) demonstrated that standard dropout, kept active at inference, approximates Bayesian inference and produces calibrated uncertainty estimates at negligible computational cost relative to ensemble methods. Yang et al. (2021) applied Bayesian PINNs to stochastic differential equations. The Monte Carlo Dropout approach used here follows Gal and Ghahramani directly, extended to the multi-domain

setting.

Multi-task PINN architectures have been explored in the context of coupled multi-physics systems (e.g., Mishra and Molinaro, 2022), where separate network heads for each physical quantity reduce inter-task gradient interference. The dual-head architecture introduced in this work applies this principle specifically to the coupled Saint-Venant system, where depth h and velocity u have fundamentally different spatial frequencies and convergence timescales.

3. Methods

3.1 The PDEModule abstraction

The central architectural contribution is the separation of the neural network engine from the physics residual. A PDEModule is an abstract class specifying four attributes: the input dimension (e.g., 2 for a (x, t) domain), the output dimension, an output transformation enforcing physical constraints (e.g., softplus to ensure depth h > 0), and a residual function that takes the network's output and computes the PDE violation via automatic differentiation.

A PINNEngine wraps any PDEModule with a shared MLP backbone using tanh activations. Tanh is required over ReLU for PINN applications because the PDE residual involves second (and in the beam case, fourth) derivatives of the network output with respect to its inputs. ReLU's piecewise-linear structure makes these higher derivatives identically zero almost everywhere. The universal training function train_pinn() operates identically regardless of which PDEModule is registered.

3.2 Governing equations

Domain	Governing equation	Order	Ground truth
Heat diffusion	$dT/dt = a\, d^2T/dx^2$	2nd parabolic	Fourier series (exact)
Wave propagation	$d^2u/dt^2 = c^2\, d^2u/dx^2$	2nd hyperbolic	Standing-wave solution (exact)
Orbital mechanics	$d^2r/dt^2 = -GM\, r / \|r\|^3$	ODE system	Kepler equation (Newton-Raphson)
Beam bending	$EI\, d^4v/dx^4 = q(x)$	4th spatial	Textbook closed-form
Open-channel flow	Saint-Venant (continuity + momentum)	1st coupled	Lax-Friedrichs FD solver

Table 1. Five physical domains, their governing equations, and analytical ground-truth sources.

3.3 Ground-truth verification

Every analytical solver was independently verified before being used to generate training data. The beam deflection formula — $v(x) = q/(24EI)(x^4 - 2Lx^3 + L^3x)$ for a simply-supported beam under uniform load — was checked against the textbook maximum deflection $v_{max} = 5qL^4/(384EI)$, confirming agreement to six decimal places. The maximum deflection also occurs at the midpoint, confirming symmetry. The

numerical fourth derivative, computed via finite differences on a 1000-point grid, equalled the applied load q/EI to within 10^{-6} relative error.

The Kepler orbit solver was verified via two conservation laws not used in its derivation. Specific angular momentum L = x(dy/dt) - y(dx/dt) was constant to a relative variation of 1.15×10^{-6} across a full orbital period. The vis-viva energy equation $v^2 = GM(2/r - 1/a)$ was satisfied with maximum relative error of 9.16×10^{-6}. Both checks used velocities obtained by numerical differentiation of the position series, not from the derivation itself.

The Saint-Venant solver uses a Lax-Friedrichs finite-difference scheme with an important correction relative to the naive explicit implementation. The friction term $S_f = n^2 u|u|/h^{4/3}$ is numerically stiff: at the timesteps this scheme actually requires for stability (set by the CFL condition), the explicit friction update overshot the discharge magnitude by a factor of 7.4 in test cases, flipping velocity sign every step. This produced clipping artefacts on 95% of grid points. The fix — treating friction semi-implicitly by solving for the updated velocity analytically — eliminated all clipping and produced physically sane velocity ranges (0.50 to 3.78 m/s for the Volta River configuration).

3.4 Dual-head architecture for Saint-Venant

An important finding emerged during development: a single-output-layer network sharing weights across both h and u plateaued at R^2 approximately 0.37 for the depth prediction regardless of training duration or data fraction. An ablation revealed the cause. A dedicated MLP trained on h alone reached $R^2 = 0.995$ at 1200 epochs, confirming that the data was entirely learnable. The plateau was therefore not a capacity or sparsity problem but a gradient-conflict problem: the Saint-Venant momentum residual, which involves nonlinear coupling of h and u through the friction slope, pushed the shared output layer in directions that benefited momentum constraint compliance at the cost of depth accuracy.

The architectural fix is a shared encoder trunk mapping (x, t) to a feature vector, followed by two separate decoder heads — one for h, one for u. The physics residual still couples both heads through the full Saint-Venant system. The per-output gradient conflict in the final layer is eliminated. Combined with the adaptive loss weighting scheme described below, the dual-head architecture raised the effective data-only ceiling for the joint prediction to $R^2 = 0.47$.

3.5 Adaptive loss balancing

Following Wang, Teng, and Perdikaris (2021), the physics weight lambda is adapted at each training step rather than fixed in advance. The gradient norms of the data loss and physics loss with respect to network parameters are computed separately every 20 training steps (computing every step approximately doubles training cost for negligible accuracy benefit). The adaptive weight is:

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lambda_new = mean(|grad_theta L_data|) / (mean(|grad_theta
                    L_pde|) + epsilon)
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An exponential moving average with coefficient alpha = 0.9 smooths the update to prevent oscillation. For the first 40% of training epochs (the warmup phase), lambda is held at zero and the network trains on data only. This warmup is essential for the Saint-Venant system: introducing the physics constraint before the network has a reasonable initial fit causes both losses to be simultaneously large, creating

conflicting gradient directions that slow convergence substantially.

3.6 Monte Carlo Dropout uncertainty quantification

Dropout layers with rate p = 0.08 are inserted after each hidden layer of the MLP. Critically, dropout is kept active at inference time by calling model.train() rather than model.eval() before each forward pass. N = 100 stochastic forward passes through the same input produce N different predictions; their mean is the point estimate and their standard deviation is the epistemic uncertainty. This procedure, due to Gal and Ghahramani (2016), approximates sampling from the posterior over network weights at negligible cost relative to training an ensemble. Wide confidence intervals identify spatial and temporal regions where additional sensor placement would most improve predictions.

3.7 Sparsity benchmark design

A VanillaMLP class provides the scientific control: identical architecture (hidden dimension 128, depth 4, tanh activations) and identical training hyperparameters, but with the physics loss weight set to zero. Any performance difference between VanillaMLP and PINNEngine at the same data fraction is attributable solely to the physics constraint. Both models are trained at data fractions {5%, 10%, 20%, 50%, 100%} with multiple random seeds per fraction. R^2 is evaluated on the full dataset in each case — measuring generalisation, not memorisation of training points.

4. Results

4.1 Domain-specific verification

All five domains produced predictions consistent with their analytical ground truths at sufficient training budget. The cleanest results appear in the 1D and ODE-governed domains.

Domain	**Data fraction**	**PINN R^2**	**Verification**
Heat diffusion	10%	0.998	PDE residual < 10^{-4} relative
Wave propagation	10%	0.994	PDE residual < 10^{-4} relative
Beam elasticity	15%	0.9998	v(L/2) = $PL^3/48EI$ exact
Orbital mechanics (x)	18%	0.872	Angular momentum: var/mean = 1.15e-6
Orbital mechanics (y)	18%	0.443	See Section 5.1 for analysis
Open-channel h (PINN)	20%	~0.37	Physics constraint active, see Section 5.2
Open-channel h (data-only)	20%	0.47	Data-only ceiling, no physics constraint
Open-channel u (PINN)	20%	0.955	Velocity prediction strong
Dam reservoir level Z(t)	All time	0.976	RK45 cross-check: 0.18% deviation

Table 2. PINN prediction accuracy per domain. R^2 measured on the full domain, not training points alone.

4.2 Beam elasticity — 4th order spatial residual

The Euler-Bernoulli beam domain provided the most striking demonstration of the physics constraint's value. Trained on only 15 out of 100 spatial sensor locations (15% of the beam), the PINN reconstructed the full deflection curve at $R^2 = 0.9998$. The analytical maximum deflection for a simply-supported beam under uniform load q is $5qL^4/(384EI) = 0.013021$ at L/2. The PINN prediction at midspan was 0.013019 — a relative error of 0.015%, indistinguishable from numerical precision. For a central point load, the piecewise solution $PL^3/(48EI)$ was also reproduced exactly.

This result is significant beyond the metric itself. The 4th-order spatial derivative chain in the residual — obtained by differentiating the network output four times with respect to x using automatic differentiation — is a genuine stress test of the framework. If autograd were not correctly chained through the tanh activations, the residual would be numerically zero, and the result would degrade to a data-only interpolation. The strong R^2 confirms that autograd flows correctly through all four derivative levels.

4.3 Orbital mechanics — sparse trajectory reconstruction

The orbital mechanics domain trained on 72 of 400 time points (18% of the trajectory) of a Keplerian orbit with eccentricity e = 0.45. The PINN recovered the x-coordinate trajectory at $R^2 = 0.872$. Crucially, the network never received the gravitational constant GM or the orbital period as an input — Newton's gravitational law $d^2r/dt^2 = -GMr/|r|^3$ was encoded as the ODE residual, and the network learned to satisfy it across the unobserved 82% of the orbit.

The R^2 for the y-coordinate was 0.443 in this run. This asymmetry was investigated across four independent training runs and found to be reproducible (R^2_x consistently 0.87-0.90, R^2_y consistently 0.40-0.45). The RMSE for y was 2.4 times larger than for x relative to each coordinate's range — confirming this is a genuine prediction quality asymmetry, not a variance-scaling artefact of the R^2 metric. The root cause is an open question. The residual formulation is symmetric in x and y; no obvious asymmetry in the training data split was identified. This is reported as a known limitation rather than papered over.

4.4 Saint-Venant — adaptive loss balancing outcome

The adaptive lambda scheme produced self-calibrated physics weights that converged to values in the range 0.08 to 1.04 depending on training stage, compared to the fixed lambda = 0.1 used in the baseline. The key diagnostic finding was that the physics constraint, rather than helping the depth prediction, actively reduced R^2 for h relative to data-only training (0.37 with physics versus 0.47 without). This was traced experimentally: a separate MLP trained on h alone reached 0.995, while the joint (h, u) fit reached 0.47 and the PINN reached 0.37. The physics constraint in this stiff, coupled system adds a gradient force that competes with the h data gradient, and even adaptive weighting cannot fully resolve this competition within the tested training budget.

The velocity u prediction, by contrast, achieved $R^2 = 0.955$ with the PINN. This asymmetry arises because u is more directly constrained by the continuity equation (which is first-order and relatively smooth) while h requires the full nonlinear momentum equation including the stiff friction slope term.

The practical implication: for this particular Saint-Venant configuration, the PINN should be used primarily for velocity prediction, and the depth prediction should be accompanied by uncertainty quantification to flag its lower reliability.

4.5 Dam and reservoir modelling

The reservoir mass balance ODE was cross-validated against a SciPy RK45 solver with tolerances of 10^{-8}. The forward-Euler implementation used in data generation deviated by 0.18% in reservoir level relative to the high-order reference — a normal discretisation error that establishes the data generation is physically correct. The PINN trained on this data achieved R^2 = 0.976 for reservoir level Z(t), demonstrating that the coupled ODE-PDE system (mass balance in the reservoir, Saint-Venant in the downstream reach) can be learned jointly by a composite two-network model.

4.6 Uncertainty quantification

Monte Carlo Dropout uncertainty bands were generated across all five domains with N = 100 stochastic forward passes per input point. For the 1D domains (beam, orbital mechanics), confidence intervals were widest in the regions farthest from training observations, consistent with genuine epistemic uncertainty about the unobserved portions of the solution. For the 2D field domains (heat, wave, fluids), uncertainty was displayed as a surface where height encodes 95% confidence interval width. The key invariant was verified by test: calling model.eval() at inference silently disables dropout and returns zero uncertainty everywhere; the correct procedure is model.train() during inference to keep dropout active.

5. Discussion

5.1 The orbital x/y asymmetry as an open problem

The persistent asymmetry between x and y reconstruction in the orbital mechanics domain warrants explicit discussion rather than a footnote. Four independent training runs with different random seeds produced R^2_x in [0.87, 0.90] and R^2_y in [0.40, 0.45]. The residual code is symmetric in x and y; there is no obvious bug. The training data covers the y-range fully. The asymmetry persists even for a circular orbit (e = 0), where x and y are mathematically equivalent (phase-shifted cosines of equal amplitude).

Several hypotheses remain untested. One is that the shared network trunk develops a bias toward the first output dimension during the early phases of training when both outputs are far from their targets. Another is that the gravitational residual for y — which involves dividing by r^3 where r is a function of both x and y — creates an asymmetric gradient path that consistently advantages x. Resolving this would require per-output gradient flow analysis, which is a worthwhile direction for future work.

5.2 Why the Saint-Venant h prediction is hard and what that means

The fluids result deserves careful interpretation for any reader considering applying PINNs to river hydrology. The data-only ceiling for the joint (h, u) prediction is R^2 approximately 0.47. Adding the Saint-Venant physics constraint reduces this to 0.37. This is not what most PINN papers report — the common narrative is that physics improves prediction in the sparse-data regime.

The difference lies in the structure of the equation. The Saint-Venant momentum equation includes a friction slope $S_f = n^2u|u|/h^{4/3}$ that is stiff (its timescale is shorter than the advection timescale) and nonlinear (it involves the absolute value of u). In the literature, PINNs have been shown to struggle specifically with stiff source terms (Krishnapriyan et al., 2021). The adaptive weighting scheme reduces but does not eliminate this problem within the tested training budget. Longer training, an implicit treatment of the friction term in the collocation residual, or a hard constraint enforcing the friction relationship as a boundary condition rather than a soft residual penalty may produce better results — each is a legitimate research direction.

The practical implication for applications such as the Volta River modelling problem that motivated this work: the PINN is reliable for velocity prediction ($R^2 = 0.955$) and for capturing the general structure of the depth field, but depth predictions should be accompanied by the Monte Carlo Dropout uncertainty bands generated in Section 4.6 rather than reported as point estimates.

5.3 The case for domain-agnostic infrastructure

A less obvious contribution of this work is the infrastructure itself. All five physics domains share one training function, one evaluation framework, one Dash application, and one test suite. Adding a new domain requires writing one PDEModule subclass — typically 40 to 60 lines of code specifying the residual function. This modularity makes the framework genuinely extensible. Future domains under consideration include Maxwell's equations for electromagnetic field reconstruction, the advection-diffusion equation for pollutant transport (directly applicable to water quality monitoring on the Volta), and the two-dimensional shallow-water equations for flood inundation mapping.

5.4 Limitations and honest reporting

Several limitations are worth stating plainly. First, all results use synthetically generated ground truth rather than real measurement data. The synthetic data is generated from verified physics solvers (checked against analytical solutions and conservation laws), but it does not capture real-world noise structures, missing data patterns, or model error from simplified governing equations. Validation against GRDC gauge station data for the Volta River is a necessary next step before any hydrological application.

Second, the sparsity benchmark in this paper uses normalised R^2 computed on normalised targets. This means the reported R^2 values measure fit quality in the normalised space. For practical engineering use, RMSE in physical units (metres, metres per second, degrees Celsius) should be the primary metric.

Third, the x/y asymmetry in orbital mechanics, the depth prediction ceiling in Saint-Venant, and the training instability noted at very small data fractions (below 5%) are all unresolved. They are reported here because hiding them would misrepresent the state of the work.

6. Conclusion

This paper presented a modular, domain-agnostic framework for physics-informed neural networks and applied it to five distinct PDE systems. The framework's central contribution is architectural: a PDEModule abstraction that cleanly separates the network engine from the physics residual, enabling any governing equation to be registered as a self-contained class and immediately supported by adaptive

training, uncertainty quantification, and systematic benchmarking.

Across the domains studied, the clearest demonstrations of the physics constraint's value appear at high sparsity. The beam elasticity domain recovered a full deflection curve from 15% of sensor readings at R^2 = 0.9998. The orbital mechanics domain reconstructed 82% of an unseen trajectory from 18% of observations, using Newton's law to extrapolate where no data existed. These are precisely the regimes — sparse sensors, expensive measurements, data-sparse regions of the world — where purely data-driven methods fail and where physics-informed approaches offer the most practical value.

The Saint-Venant result is a counterexample that is equally informative. The physics constraint, implemented as a soft residual penalty, can actively degrade prediction accuracy when the governing equation is stiff and its gradient conflicts with the data gradient. This is a known issue in the PINN literature but is rarely demonstrated as clearly as in the ablation studies here. The adaptive loss balancing scheme reduces but does not eliminate the problem. Stiff friction terms in open-channel flow remain a genuine open problem for physics-informed machine learning.

The complete codebase, interactive application, and standalone orbital visualisation are publicly available. The framework is designed for extension: adding a new physical domain requires one subclass and approximately 50 lines of code. Planned extensions include two-dimensional flood inundation mapping, pollutant transport coupled to the river flow field, and integration with GRDC hydrological gauge station data for the Volta River basin.

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